

Abstract

Wireless Sensor Networks (WSNs) deployed in disaster-monitoring scenarios require energy-efficient, low-latency multi-hop routing under strict resource constraints. We propose **Graph-Attentive Policy Fusion (GAPF)**, a lightweight meta-learning framework that fuses frozen MARL expert policies (QMIX, QTRAN) through a **Graph Convolutional Network (GCN)** encoder and a **Contextual Algorithm Selection (CAS)** controller. GAPF encodes the sensor topology into a compact graph embedding and selects the best-performing expert with only **1,473 trainable parameters**. Evaluated across 7 scenarios (20–100 sensors, 21,000 test episodes), GAPF achieves $\geq 98\%$ packet delivery, $+43\%$ to $+111\%$ improvement over baselines, $2\text{--}3.5\times$ less energy, $3\text{--}9\times$ better energy balance, and $13\times$ less memory.

Keywords: WSN, MARL, GNN, policy fusion, energy-efficient routing, disaster monitoring.

Introduction & Motivation

- Disaster-monitoring WSNs require **reliable, energy-efficient multi-hop routing** to a base station under strict resource constraints.
- Existing MARL approaches (QMIX, QTRAN) treat routing independently, ignoring the network topology, and suffer from **memory explosion at scale** (up to 13.8 GB).
- Metaheuristic baselines (e.g. QPSOFL) see packet delivery **collapse below 10%** in large networks.
- Goal:** Design a lightweight, topology-aware meta-controller that selects the best routing expert per deployment while satisfying energy, latency, and memory constraints.

Problem Formulation

Setting: Base station (BS) at the centre of a 100×100 m field + N sensor nodes. Each sensor starts with $E_0 = 1$ J and holds 1 packet. Multi-hop relay to BS within radius R .

Energy Model

$$\begin{cases} E_T(M, d) = M \cdot a \cdot (E_{\text{elec}} + E_{\text{amp}} \cdot d^2) \\ E_R(M) = M \cdot a \cdot E_{\text{elec}} \end{cases}$$

with $E_{\text{elec}} = 50$ nJ/bit, $E_{\text{amp}} = 100$ pJ/(bit·m²), packet size $a = 512$ bits.

Routing Objectives

- Maximize** packet delivery ratio (PDR).
- Minimize** total energy consumption $\sum (E_T + E_R)$.
- Balance** energy usage across sensors (minimize $\sigma(e_{\text{rem}})$).

Reward Scalarization

4-dimensional reward vector per relay hop:

$$\mathbf{r}_i^{(t)} = \left[\underbrace{r_1}_{\text{angle}}, \underbrace{r_2}_{\text{energy}}, \underbrace{r_3}_{\text{balance}}, \underbrace{r_4}_{\text{load}} \right]$$

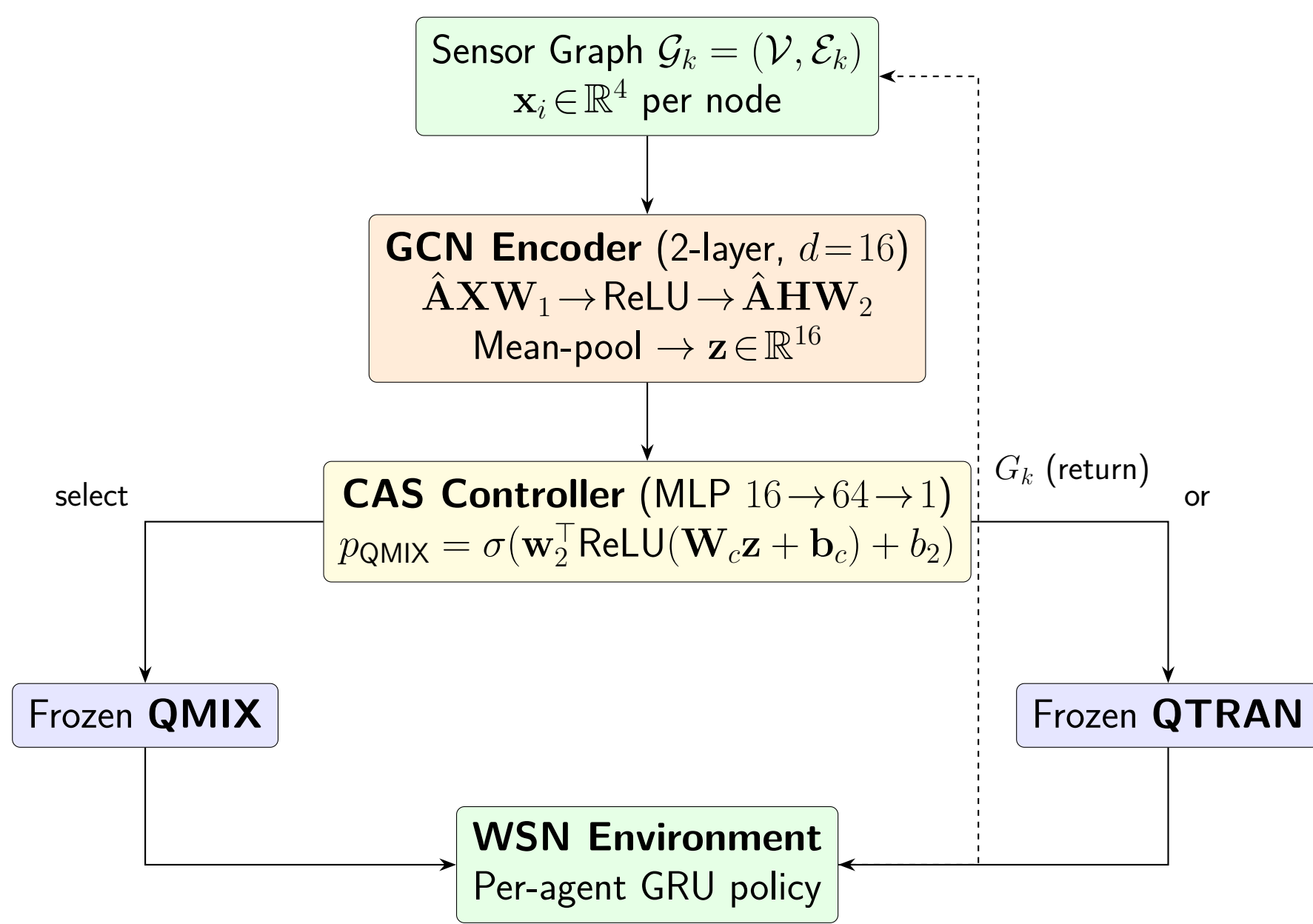
Monotonic Q-K Attention Network scalarizes $\mathbf{r}$:

$$\alpha_j = \frac{\exp(\mathbf{k}_j^\top \mathbf{q} / \sqrt{d})}{\sum_{l=1}^4 \exp(\mathbf{k}_l^\top \mathbf{q} / \sqrt{d})}, \quad g_\psi(\mathbf{r}') = \mathbf{W}_3 \sigma(\mathbf{W}_2 \sigma(\mathbf{W}_1 \mathbf{r}' + \mathbf{b}_1) + \mathbf{b}_2) + b_3$$

$\sigma(\cdot)$ = softplus, all $\mathbf{W}_\ell \geq 0$ (monotonicity), dims $4 \rightarrow 16 \rightarrow 16 \rightarrow 1$.

Outer loop: Evolution Strategies (ES) tunes attention parameters to maximize J = energy-eff. + energy-balance $\in [0, 2]$.

GAPF Architecture



Trainable Parameters

$$|\theta| = \underbrace{4 \times 16 + 16 \times 16}_{\text{GCN: 320}} + \underbrace{16 \times 64 + 64 + 64 \times 1 + 1}_{\text{CAS: 1,153}} = \mathbf{1,473}$$

$\sim 26\times$ smaller than a single RNN agent (38,983 params). Trained via **REINFORCE** with EMA baseline ($\beta=0.95$).

Design Rationale

- Frozen experts** — avoids catastrophic forgetting; only the meta-controller is trained.
- GCN encoder** — captures topology (node degree, proximity to BS); critical in sparse networks.
- Episode-level selection** — one forward pass of GCN+CAS per episode (~ 0.03 ms).
- CTDE deployment** — GCN+CAS on gateway (6 KB); each sensor runs a GRU (~ 100 KB flash, ~ 2 KB SRAM).
- ES outer loop** — derivative-free tuning of reward attention; robust to non-differentiable objectives.

Evaluation Scenarios

#	Sensors (n)	Radius (R)	Rationale
1	20	25 m	Sparse, short range — extreme challenge
2	30	25 m	Small–medium, limited relay options
3	50	35 m	Medium density, standard range
4	70	35 m	Dense baseline configuration
5	70	45 m	Dense, extended range
6	100	35 m	Large-scale scalability test
7	100	50 m	Large-scale, high connectivity

3 seeds per scenario $\times$ 1,000 episodes = **21,000 test episodes** total.

Packet Delivery Ratio (PDR)

n	R	QMIX	QTRAN	QPSOFL	GAPF
20	25	.559 $\pm$.257	.558 $\pm$.258	.075 $\pm$.107	.753 $\pm$.161
30	25	.568 $\pm$.179	.567 $\pm$.174	.053 $\pm$.046	.877 $\pm$.159
50	35	.553 $\pm$.248	.552 $\pm$.250	.153 $\pm$.092	.990 $\pm$.088
70	35	.545 $\pm$.242	.545 $\pm$.243	.122 $\pm$.090	.980 $\pm$.119
70	45	.667 $\pm$.246	.659 $\pm$.256	.365 $\pm$.092	.980 $\pm$.124
100	35	.464 $\pm$.270	.462 $\pm$.261	.099 $\pm$.080	.980 $\pm$.124
100	50	.669 $\pm$.266	.684 $\pm$.270	.448 $\pm$.057	.980 $\pm$.124

Energy Balance (σ_E , lower is better)

n	R	QMIX	QTRAN	QPSOFL	GAPF
20	25	.0075	.0075	.0056	.0043
30	25	.0111	.0112	.0085	.0035
50	35	.0264	.0265	.0081	.0039
70	35	.0318	.0319	.0114	.0049
70	45	.0379	.0382	.0024	.0061
100	35	.0455	.0457	.0144	.0061
100	50	.0559	.0559	.0010	.0061

Deployment Feasibility

Metric	QMIX	QTRAN	QPSOFL	GAPF
Peak RAM (MB)	3,160–13,757	3,160–13,757	178–195	684–951
Mem. <1,024 MB?	FAIL	FAIL	PASS	PASS
Latency p_{95} (ms)	0.27	0.17		0.03
Agent flash	~ 100 KB (shared GRU architecture)			

Key Findings

- F1** GAPF $\geq 98\%$ PDR; advantage grows in sparser networks ($+111\%$ at $n=100$, $R=35$).
- F2** $2\text{--}3.5\times$ lower energy, $3\text{--}9\times$ better balance vs QMIX/QTRAN.
- F3** $\text{QMIX} \approx \text{QTRAN}$ ($<2\%$ gap): gain comes from *topology-aware fusion*, not expert choice alone.
- F4** QPSOFL collapses at scale; its low energy is an artifact of near-zero delivery.
- F5** GAPF is the **only algorithm passing all feasibility constraints** (memory, latency).

Conclusion

GAPF demonstrates that lightweight policy fusion with graph-structural awareness can replace heavyweight monolithic MARL for resource-constrained, safety-critical WSN routing. With only 1,473 parameters it achieves near-perfect delivery, $2\text{--}3.5\times$ less energy, and feasible deployment on embedded hardware (ARM Cortex-M4, ~ 100 KB flash).

Future work: hardware-in-the-loop validation on nRF52840 motes; cross-domain disaster monitoring (flood/wildfire/earthquake); dynamic step-level expert fusion.

References

- [1] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," *arXiv preprint arXiv:1609.02907*, 2017.
- [2] R. J. Williams, "Simple statistical gradient-following algorithms for connectionist reinforcement learning," *Machine Learning*, vol. 8, no. 3–4, pp. 229–256, 1992.
- [3] T. Salimans, J. Ho, X. Chen, S. Sidor, and I. Sutskever, "Evolution strategies as a scalable alternative to reinforcement learning," *arXiv preprint arXiv:1703.03864*, 2017.
- [4] T. Rashid, M. Samvelyan, C. Schroeder de Witt, G. Farquhar, J. Foerster, and S. Whiteson, "Monotonic value function factorisation for deep multi-agent reinforcement learning," *Journal of Machine Learning Research*, vol. 21, no. 178, pp. 1–51, 2020.
- [5] K. Son, D. Kim, W. J. Kang, D. E. Hostallero, and Y. Yi, "QTRAN: Learning to factorize with transformation for cooperative multi-agent reinforcement learning," in *Proc. ICML*, pp. 5887–5896, 2019.
- [6] X. Hu et al., "Energy-efficient routing in wireless sensor networks using quantum particle swarm optimization with fuzzy logic," *IEEE Sensors Journal*, 2024.
- [7] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [8] J. Sill, "Monotonic networks," *Advances in Neural Information Processing Systems*, vol. 10, 1997.